Wednesday, 22 January 2025 12:07 PM

Macro Economics

Books → Gregory Mankiw
Dornbush Fischer Startz } Introductory Macroeconomics

Production

$Y = F(L, K)$
labour — Capital
tradeoff

- Wage rate (w)
- Rental rate (r)

Amount paid / Cost = $wL + rK$

Output / Income / Amount Earned / Revenue = Cost + Profit
(W) (Z)

Total Money gained from the market

$\boxed{Y = wL + rK + Pr}$

My cost is income of someone else

GDP → Gross Domestic Product = $\sum Y$
→ Geographical location matters, owner not

Nike USA → Productive units China → Income generated is counted in
GDP of China
GNP of USA (Gross National Product)

GNP = GDP + net property income from abroad.

★ NDP (Net Domestic Product) →
NDP = GDP − Capital depreciation (replenishment of capital)
(~ 10–11% of GDP)

★ NI (National Income) → (~ 80% GDP)
NI = NDP − (Indirect taxes) (10% of NDP)
= GDP − Capital depreciation − Indirect tax (Income of Government)

Demand

★ Aggregate Demand →

$AD = C + \underline{I + G + (X - M)}$

X → Export
M → Import

C → Σ all the consumption demands
I → Investment component
G → Consumption or expenditure by Gov on Infrastructure...

$C = \bar{C} + cY$ (Income ↑ Consumption ↑)

$\bar{C}$ → Inheritance amount... (Baremin consumption)

cY → Proficiency of consumption

Savings ← $1-c$
$0 < c < 1$ ← Marginal propensity to consume

$Y \equiv AD$
Output ≡ Demand
$Y \equiv GDP$

→ your demand is someones' income
∴ AD drives production (Y) → GDP

→ Spending ↑ → Income ↑ → Spending ↑

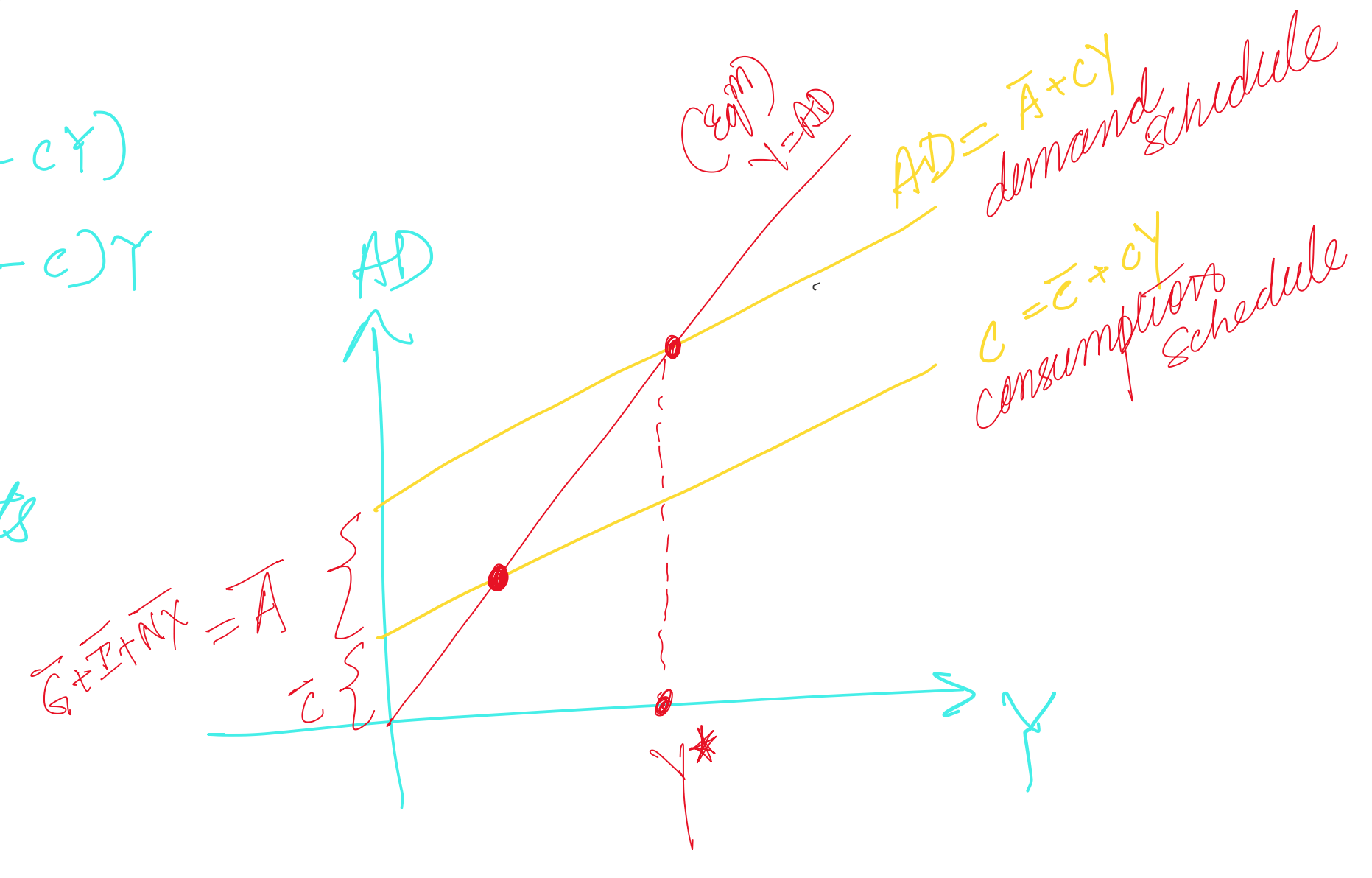
Equilibrium Situation
↓
$\underline{\underline{Y = AD}}$

→ Unplanned Inventory (Stores)
$IU = Y - AD$ (Gap b/w output demanded and produced)
↓
$IU > 0$ ⇒ Inventory investment (stocks piling)
(accumulation of stocks → cut down on production)

$IU < 0$ ⇒ Inventory disinvestment
(need to provides goods from outside the inventory)

Savings →
$S = Y - C$
$S = Y - (\bar{C} + cY)$
$S = -\bar{C} + (1-c)Y$

Thus, $AD = C + \underbrace{I + G + NX}_{\bar{A}}$ constants
$= \bar{C} + cY + \bar{A}$
$= cY + \bar{A}$



Consumption, Aggregate Demand & Autonomous Spending

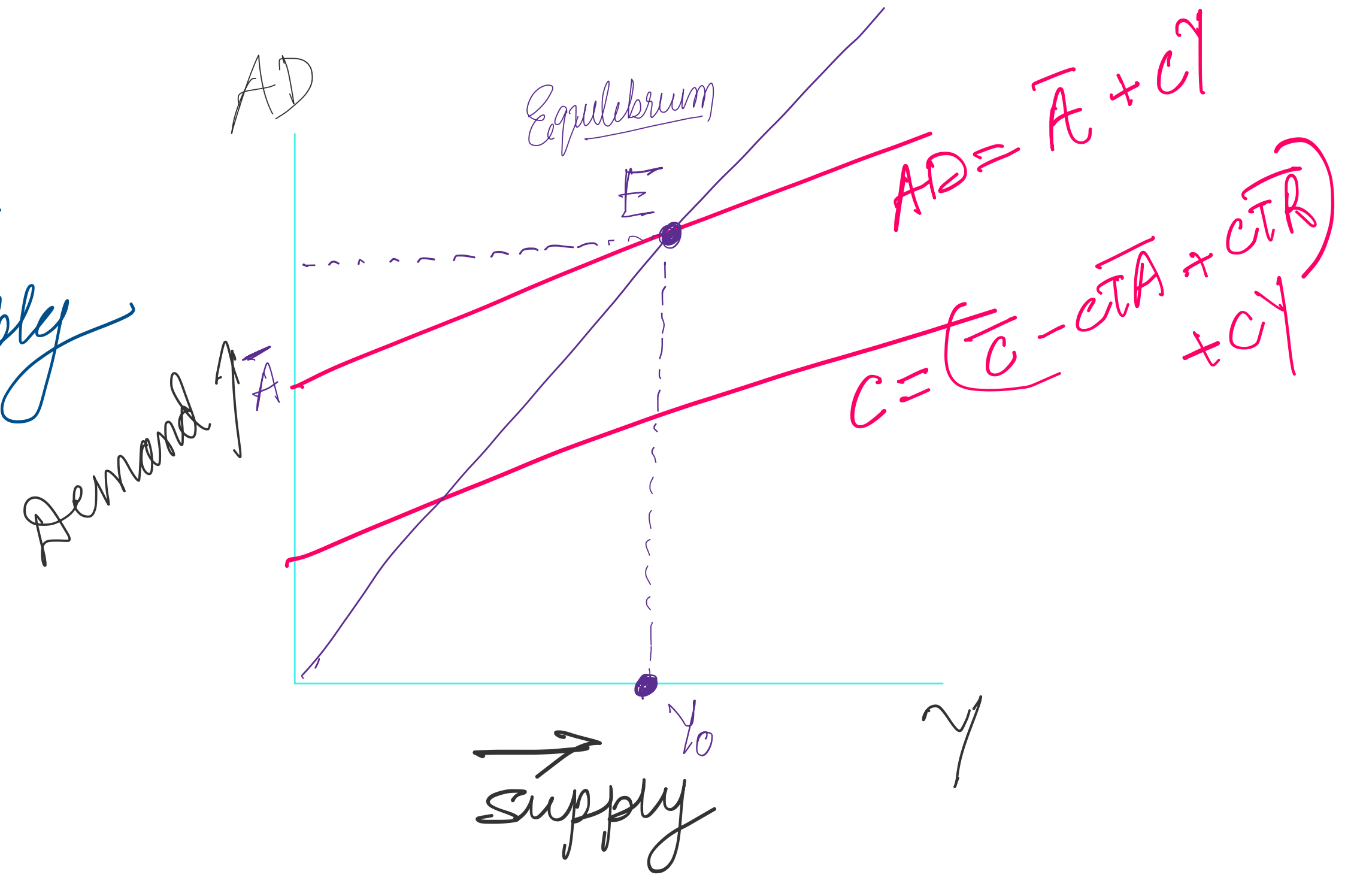
Components of AD:
Investment − $\bar{I}$
Govt Spending − $\bar{G}$
Taxes → $\overline{TA}$
Govt transfers (Subsides) → $\overline{TR}$
Net Exports (Demand by foreigners) → $\overline{NX}$
} fixed / Autonomous amounts

→ Disposable Income →
$YD = Y - \overline{TA} + \overline{TR}$
$C = \bar{C} + cYD$
$= \bar{C} + c(Y - \overline{TA} + \overline{TR})$

∴ $AD = C + I + NX + G$
$= \bar{C} + c(Y - \overline{TA} + \overline{TR}) + \bar{I} + \overline{NX} + \bar{G}$
$= (\bar{C} - c\overline{TA} + c\overline{TR} + \bar{I} + \overline{NX} + \bar{G}) + cY$

$\boxed{AD = \bar{A} + cY}$

AD → aggregate demand
cY → aggregate supply



Income Y = Consume C + Invested / Save I
$I \equiv S$

Eqm (Y=AD)
$Y = C + I + G + NX$
$YD = Y + TR - TA$
$YD = C + S$
Desposable Income

$C + S = C + I + G + NX + TR - TA$

$\boxed{S - I = (G + TR - TA) + NX}$

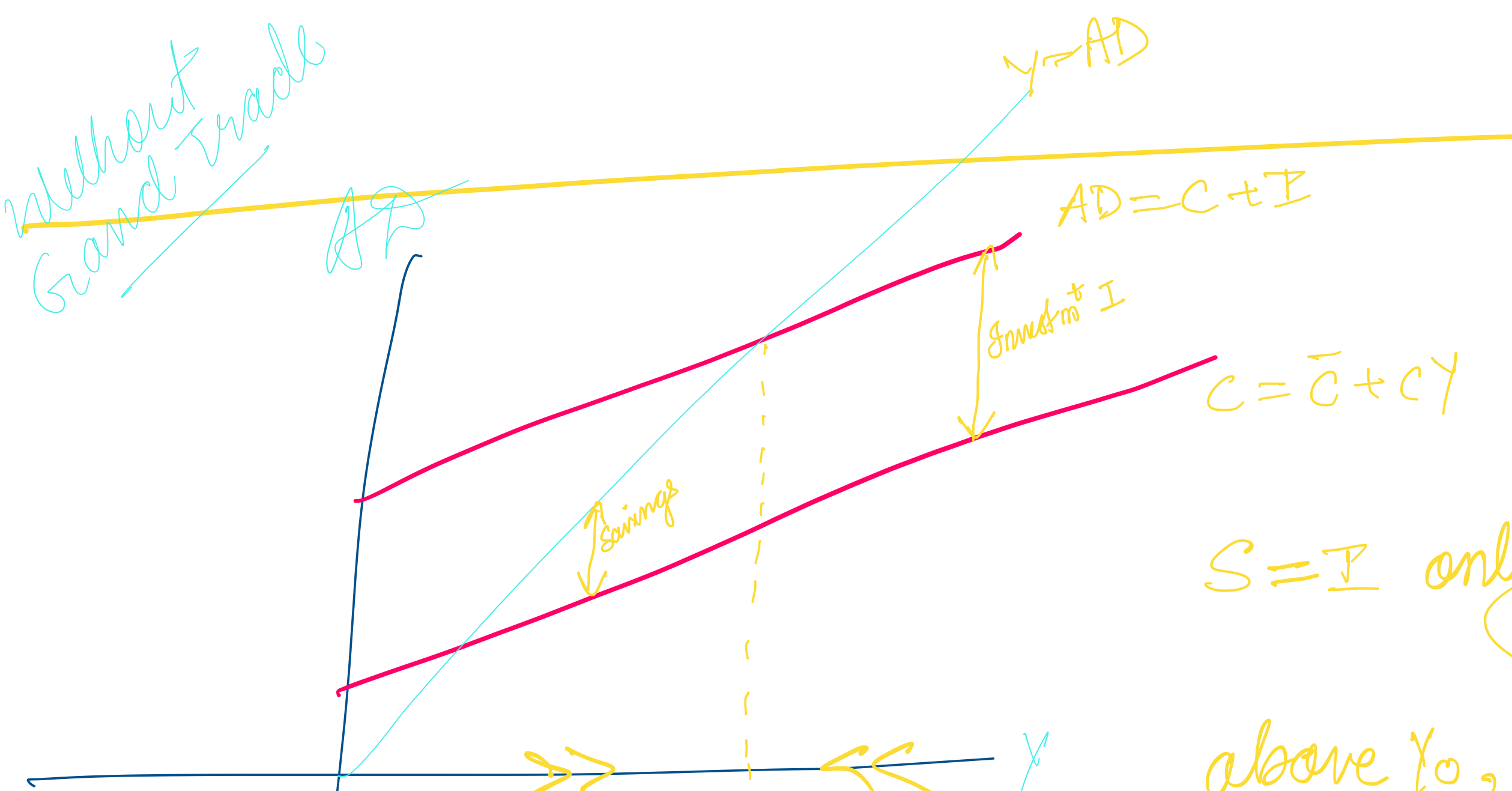
$S = I + \underbrace{(G + TR - TA)}_{\text{Buget Deficit}} + NX$

S → private saving
I → Total private Investment
(G + TR − TA) → Buget Deficit → Government
NX → Net Exports & → Trade

Government, Trade — without them → $S = I$

Rombinson-Cruso economy

The savings will be used for financing Government's Budget Deficit
(different kinds of Investment)



S=I only at Equilibribrium

above Y₀, S > I
below Y₀, S < I

$Y = C + S + TA - TR$
$Y = C + I + G + NX$
$I = S + (TA - TR - G) - NX$